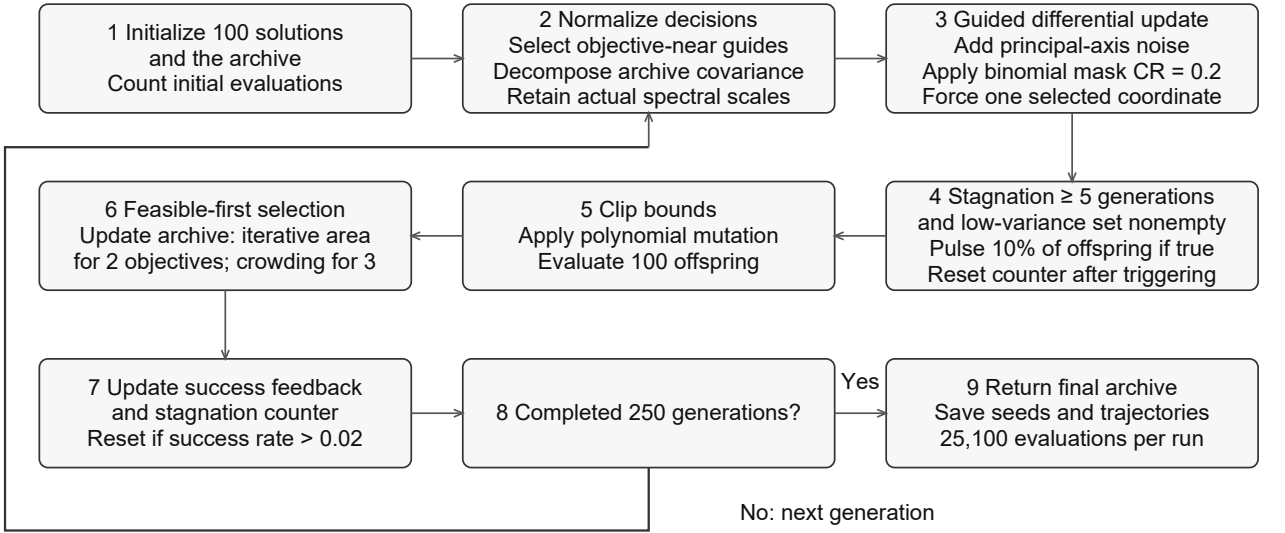
1 Initialize 100 solutions
and the archive
Count initial evaluations
2 Normalize decisions
Select objective-near guides
Decompose archive covariance
Retain actual spectral scales
3 Guided differential update
Add principal-axis noise
Apply binomial mask CR = 0.2
Force one selected coordinate
4 Stagnation ≥ 5 generations
and low-variance set nonempty
Pulse 10% of offspring if true
Reset counter after triggering
5 Clip bounds
Apply polynomial mutation
Evaluate 100 offspring
6 Feasible-first selection
Update archive: iterative area
for 2 objectives; crowding for 3
7 Update success feedback
and stagnation counter
Reset if success rate > 0.02
8 Completed 250 generations?
Yes
9 Return final archive
Save seeds and trajectories
25,100 evaluations per run
No: next generation